Two ways to simulate a tilting drone

Both update full orientation. The difference is how rotation responds to a command.

A | EXISTING CUSTOM PLANT

Simplified thrust / body-rate dynamics

Quadrotor NMPC

Reference + current state → thrust T and body rates p, q, r

↓

Approximate actuator response

Body-rate command → achieved rates through a response lag

↓

Update orientation and translation

Integrate achieved rates; thrust points along body +Z

↓

Render from the resulting camera pose

Actual position + roll / pitch / yaw + fixed camera mount

Feed the resulting state back to the next MPC decision

B | EXISTING NATIVE ISAAC PLANT

Rigid-body dynamics with inertia

Quadrotor NMPC

Reference + current state → thrust T and body rates p, q, r

↓

Body-rate tracking controller

Rate error → bounded torque; thrust also has response lag

↓

PhysX integrates forces and torques

Inertia determines rotation; tilted thrust changes translation

↓

Render from the resulting camera pose

Actual position + roll / pitch / yaw + fixed camera mount

Feed the resulting state back to the next MPC decision

YOUR PROPOSAL | change the planner, retain a quadrotor plant

Linear MPC

XYZ acceleration + yaw

→

Geometric controller

Desired tilt → thrust / rates

→

Quadrotor plant

Orientation + translation

→

Camera renderer

RGB from actual pose

Same principle in all cases: render the achieved camera pose, rather than a level pose placed on the reference path.

Conceptual architecture • The proposed linear-MPC / geometric-controller combination is not yet wired into our simulator.